

AUTOMATION-DRIVEN MICRO-SCALE SPATIAL METAGENOMICS



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BACKGROUND

Microbiota

Host-microbiome & microbe-microbe interactions are critical for the well-being & health of the host organisms.

Zoom-in

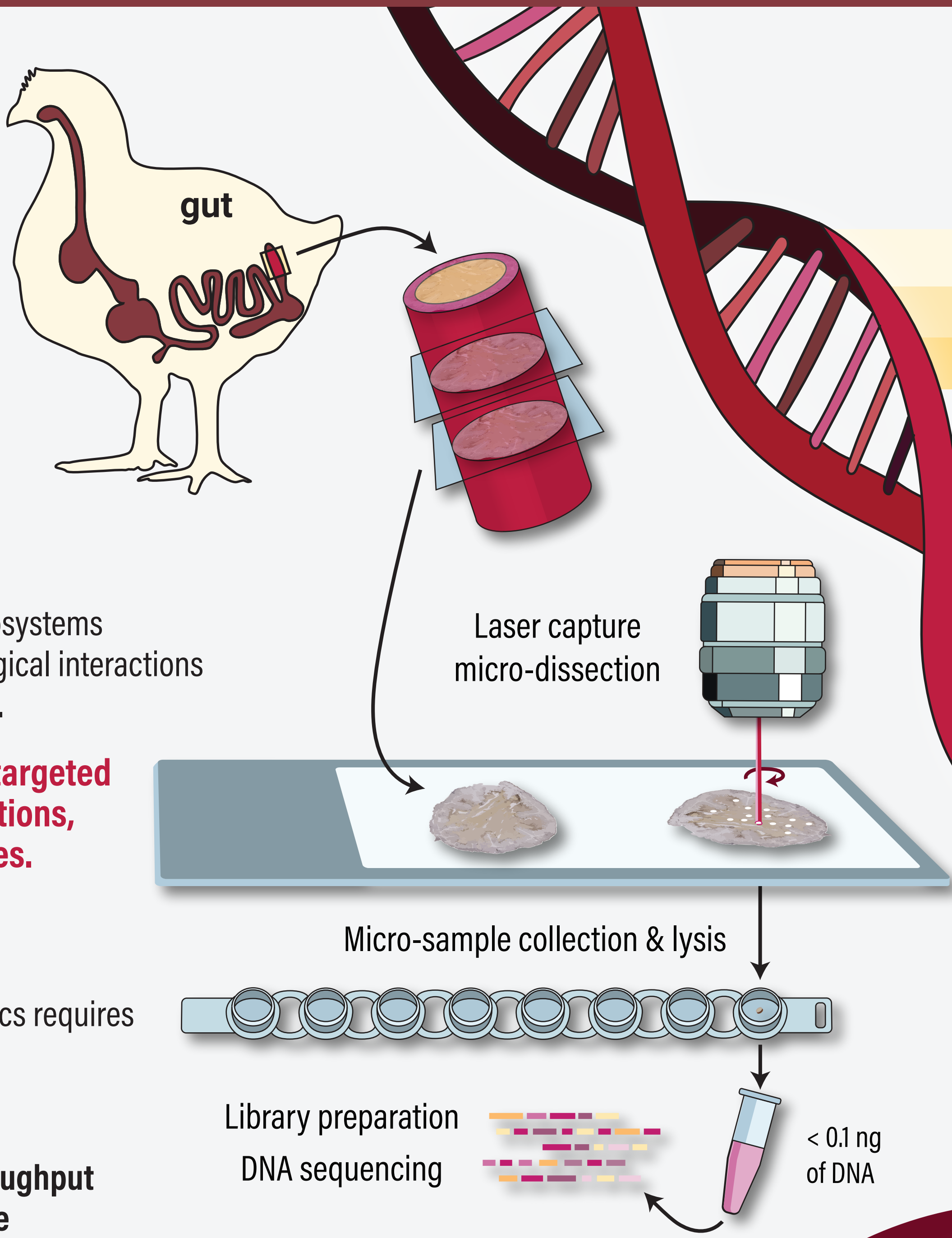
Generating 3D reconstructions of intestinal host-microbiota ecosystems we can accurately explore biological interactions between microbes & their hosts.

This is essential to develop targeted microbiota-modulation solutions, such as probiotics & vaccines.

Automation

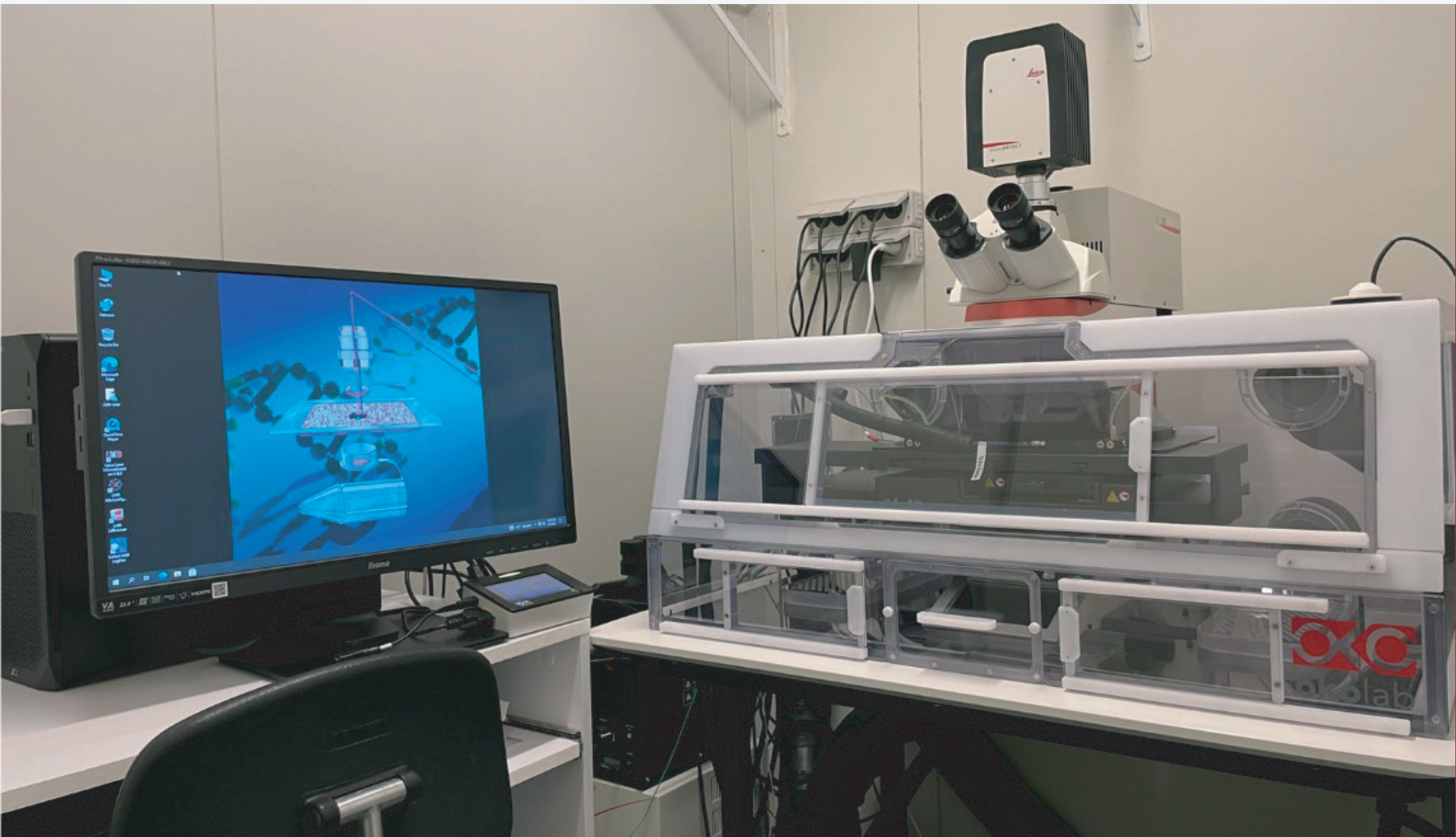
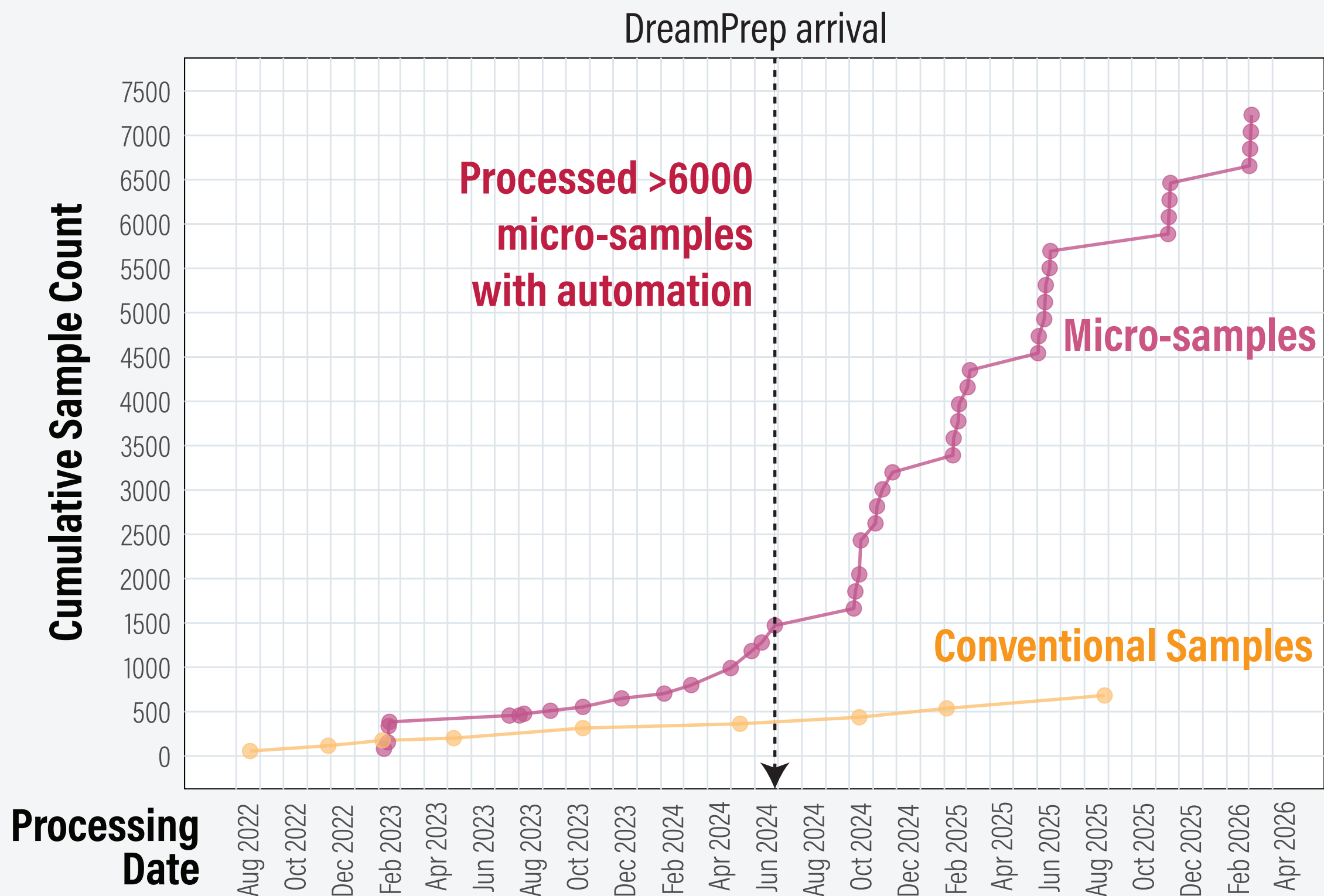
Micro-scale spatial metagenomics requires the generation of hundreds of micro-samples per host.

Automation enables high-throughput and standardised micro-sample processing.



OPTIMISED SAMPLE PROCESSING

	Time	Sustainability
Manual - 96 samples	~9h hands-on	2980 g of plastic
Automated - 96 samples	~7h hands-off	2192 g ↓ 26.4 %
Automated - 192 samples	~9h hands-off	2043 g ↓ 31.4 %



Leica LMD7

LIBRARY PREPARATION AUTOMATION

Process overview

1. Micro-dissection (Leica LMD7) & logging of coordinates
2. Overnight lysis of micro-samples
3. Library preparation (Ovation Ultralow V2 Kit & DreamPrep, Tecan)
4. Metagenomic shotgun sequencing (NovaSeq X Plus, Illumina)

Tecan Fluent, DreamPrep 780



Solved Challenges

- Micro-samples have extremely low-yield DNA: ~10-100 pg
- Restructure script to avoid unnecessary sample pipetting
- Accommodate changes in plate dimensions due to overnight incubation
- Miniaturisation of reactions improved library quality
- Adjust liquid classes for transfer of volumes as low as 2.5 µL
- 3D-reconstructions require thousands of micro-samples
- Increase through-put from 96 to 192 samples
- Accommodate additional labware on deck
- Interweave command over incubation for time efficiency

Read more about how we developed Micro-scale spatial metagenomics in our pre-print:

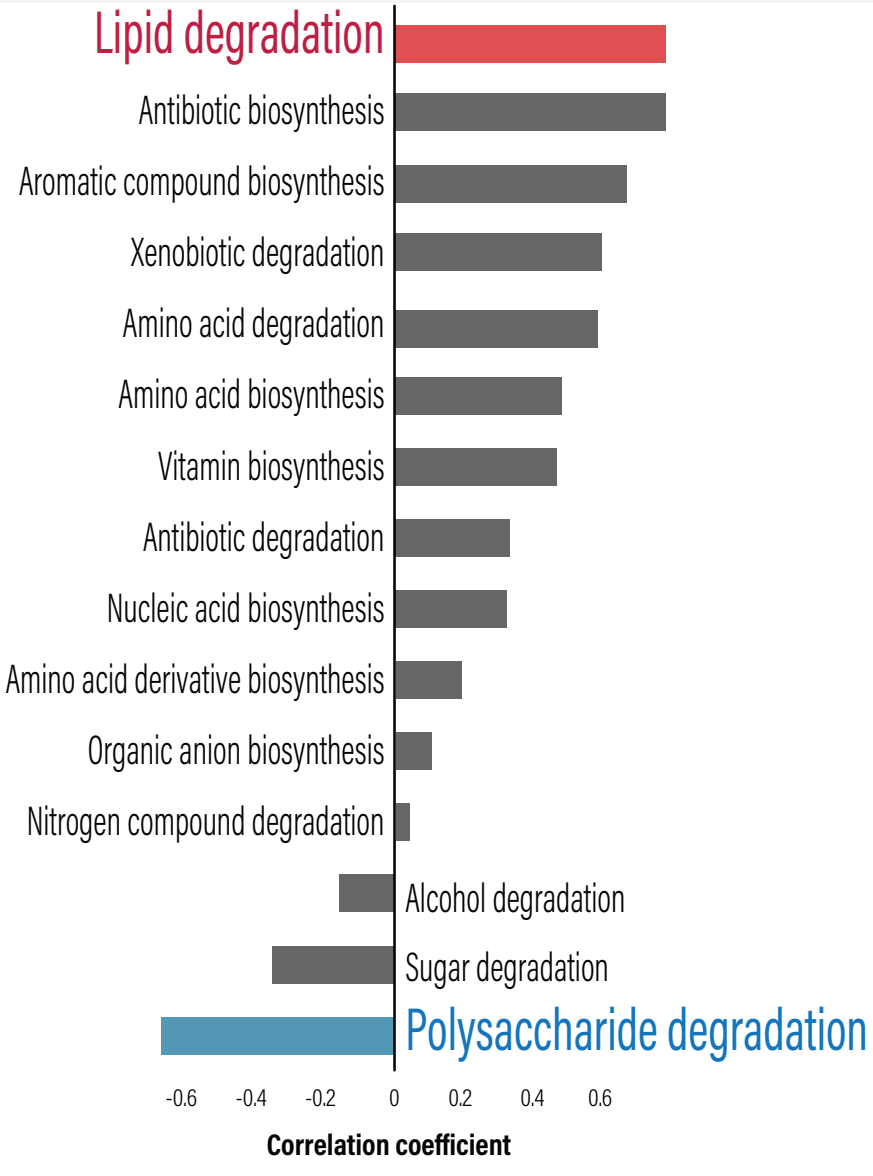
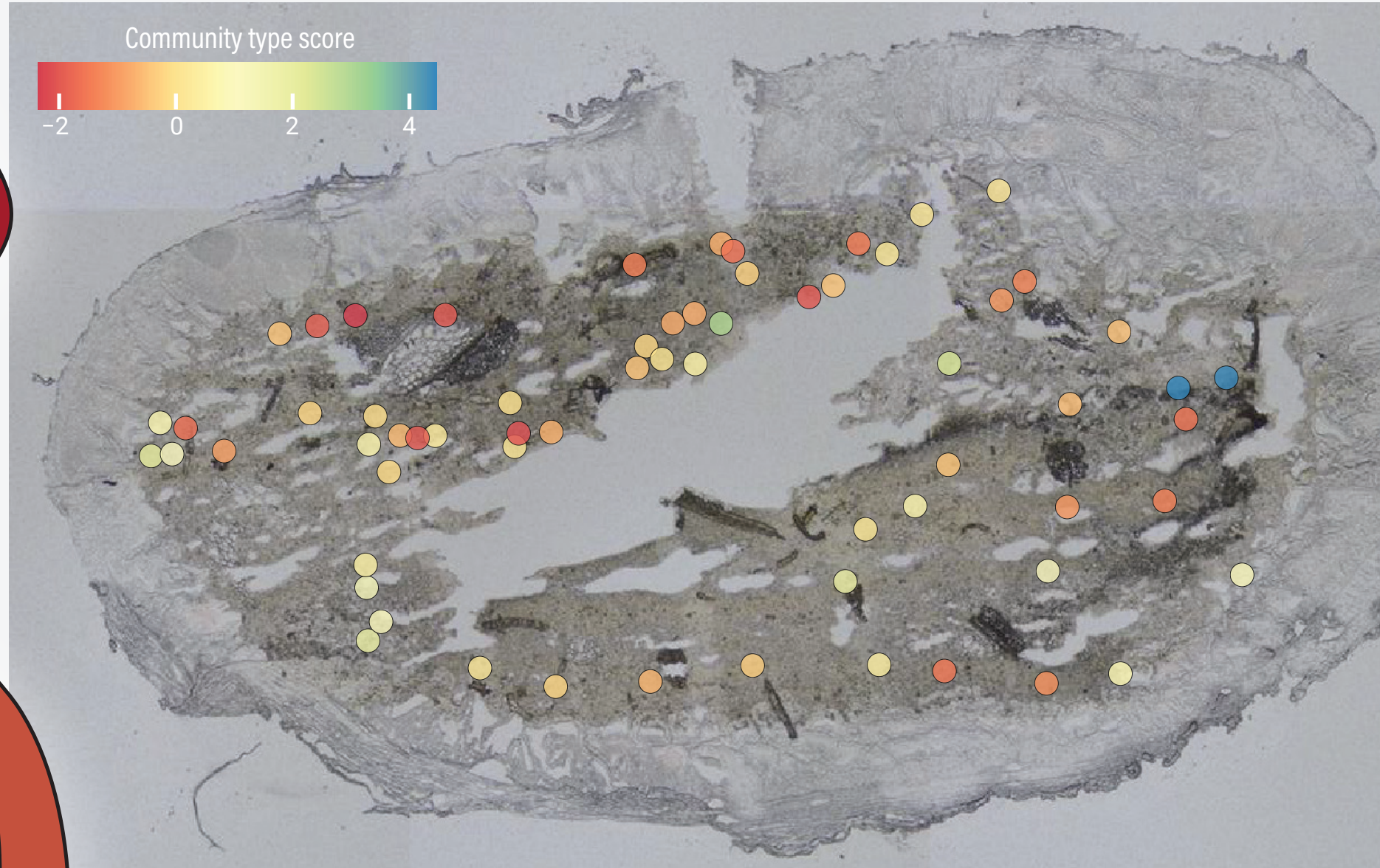
“Micro-scale spatial metagenomics: revealing high-resolution spatial biogeography of gut microbiomes”



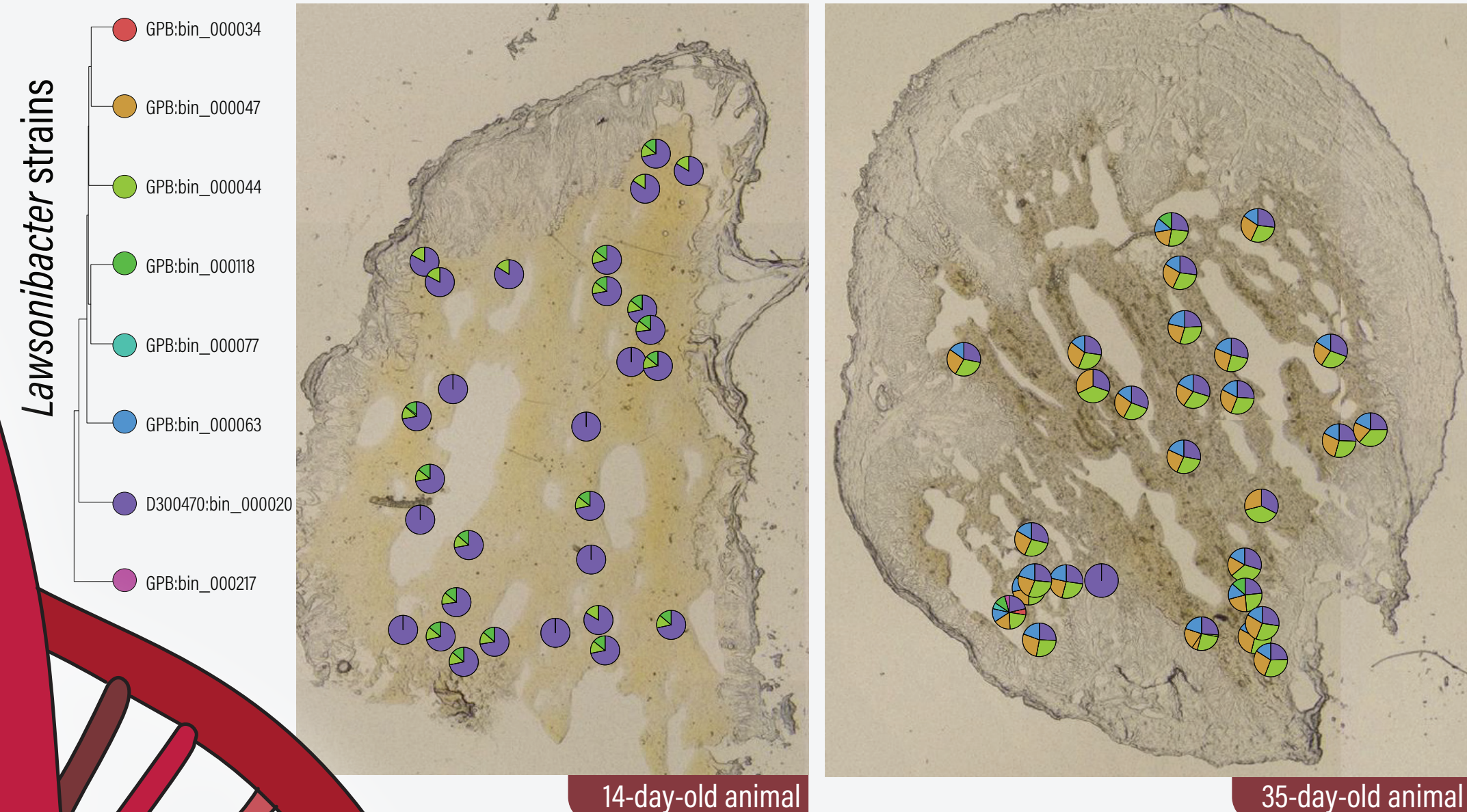
bioRxiv
THE PREPRINT SERVER FOR BIOLOGY

SELECTED RESULTS

Combined **spatial**, **phylogenetic**, and **functional** analyses of the micro-samples can detect **spatial signal**, if the gut content is spatially organised.



Strain level variations can be detected within **cryosections**, and across different **hosts**.



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